

Lab Tasks

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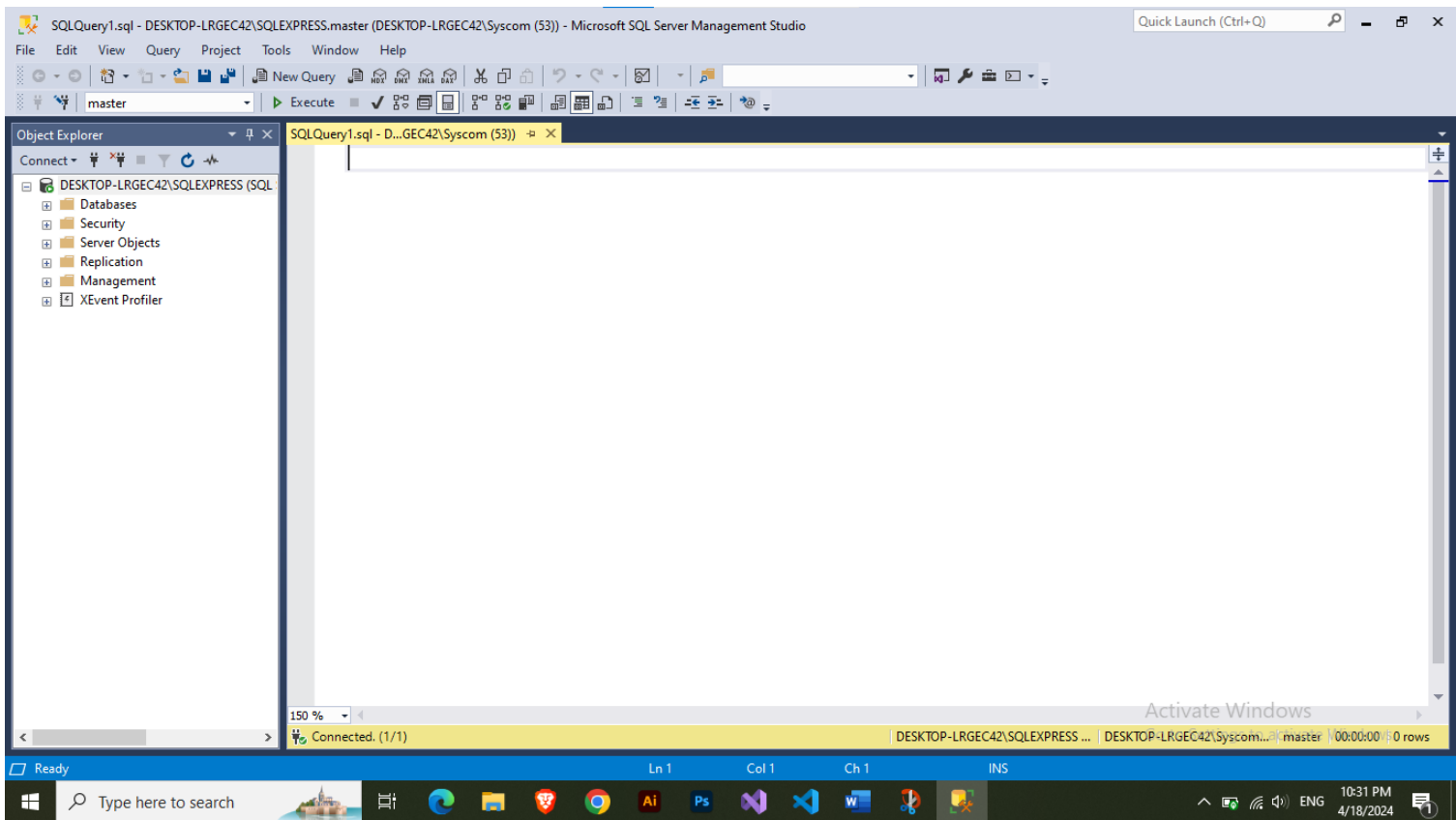
Course: Database Management Systems Lab (DBMS)

Program: (BS) - Computer Science

Lab #01

Introduction to Database and Microsoft SQL Server

Task 1: Installation of Microsoft SQL Server.



Microsoft SQL server management studio

Lab #02

Data Definition Operations and to become familiar with data definition language.**Task 1: Create a new table Person and insert at least 5 records****Code:**

```

CREATE TABLE Person(
  CNIC INT PRIMARY KEY,
  Name VARCHAR(50),
  Age INT,
  Nationality VARCHAR(50),
  Gender VARCHAR(10),
);

INSERT INTO Person VALUES(42301, 'Faisal', 20, 'Pakistani', 'Male');
INSERT INTO Person VALUES(42302, 'Bilal', 20, 'Pakistani', 'Male');
INSERT INTO Person VALUES(42303, 'Furqan', 21, 'Pakistani', 'Male');
INSERT INTO Person VALUES(42304, 'Farhan', 22, 'Pakistani', 'Male');
INSERT INTO Person VALUES(42305, 'Ahsan', 19, 'Pakistani', 'Male');

```

Results:

Results		Messages			
	CNIC	Name	Age	Nationality	Gender
1	42301	Faisal	20	Pakistani	Male
2	42302	Bilal	20	Pakistani	Male
3	42303	Furqan	21	Pakistani	Male
4	42304	Farhan	22	Pakistani	Male
5	42305	Ahsan	19	Pakistani	Male

Task 2: Create a new table Customer and insert at least 5 records.**Code:**

```

CREATE TABLE Customer (
  customer_id INT PRIMARY KEY,
  full_name VARCHAR(100),
  date_of_birth DATE,
  gender VARCHAR(10),
  address VARCHAR(255)
);

INSERT INTO Customer VALUES
(1, 'Faisal', '2003-03-03', 'Male', 'Bahdurabad, Karachi'),
(2, 'Bilal', '2004-09-20', 'Male', 'Burns Road, Karachi'),
(3, 'Furqan', '2002-03-10', 'Male', 'Bahdurabad, Karachi'),
(4, 'Farhan', '2004-11-28', 'Male', 'Saddar, Karachi'),
(5, 'Ahsan', '2000-07-03', 'Male', 'DHA, Karachi')

```

Results:

	customer_id	full_name	date_of_birth	gender	address
1	1	Faisal	2003-03-03	Male	Bahdurabad, Karachi
2	2	Bilal	2004-09-20	Male	Bums Road, Karachi
3	3	Furqan	2002-03-10	Male	Bahdurabad, Karachi
4	4	Farhan	2004-11-28	Male	Saddar, Karachi
5	5	Ahsan	2000-07-03	Male	DHA, Karachi

Task 3: Create a new table Order and insert at least 5 records**Code:**

```
CREATE TABLE Orders(
    order_id INT PRIMARY KEY,
    customer_id INT,
    order_date DATE,
    product_name VARCHAR(255),
    quantity INT,
    total_price INT,
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)
);
```

```
INSERT INTO Orders VALUES
(1, 1, '2024-03-28', 'Laptop', 1, 1200),
(2, 2, '2024-03-27', 'Smartphone', 2, 800),
(3, 3, '2024-03-26', 'Headphones', 1, 100),
(4, 1, '2024-03-25', 'Tablet', 1, 500),
(5, 4, '2024-03-24', 'Speaker', 1, 150);
```

Results:

	order_id	customer_id	order_date	product_name	quantity	total_price
1	1	1	2024-03-28	Laptop	1	1200
2	2	2	2024-03-27	Smartphone	2	800
3	3	3	2024-03-26	Headphones	1	100
4	4	1	2024-03-25	Tablet	1	500
5	5	4	2024-03-24	Speaker	1	150

Task 4: Add a new field in the Person table.**Code:**

```
ALTER TABLE Person
ADD Email VARCHAR(100);
```

Task 5: Modify any field in the Person table**Code:**

```
UPDATE Person
SET Email = 'faisal@gmail.com'
WHERE CNIC = 42301;
```

```
UPDATE Person
SET Email = 'bilal@gmail.com'
WHERE CNIC = 42302;
```

```
UPDATE Person
SET Email = 'furqan@gmail.com'
WHERE CNIC = 42303;
```

```
UPDATE Person
SET Email = 'farhan@gmail.com'
WHERE CNIC = 42304;
```

```
UPDATE Person
SET Email = 'ahsan@gmail.com'
WHERE CNIC = 42305;
```

Results:

	CNIC	Name	Age	Nationality	Gender	Email
1	42301	Faisal	20	Pakistani	Male	faisal@gmail.com
2	42302	Bilal	20	Pakistani	Male	bilal@gmail.com
3	42303	Furqan	21	Pakistani	Male	furqan@gmail.com
4	42304	Farhan	22	Pakistani	Male	farhan@gmail.com
5	42305	Ahsan	19	Pakistani	Male	ahsan@gmail.com

Task 6: Use some constraints in your queries.

Code:

```
CREATE TABLE Employee (
    emp_num INT,
    emp_name VARCHAR(10) NOT NULL,
    dept_num INT NOT NULL,
    email VARCHAR(50),
    CONSTRAINT emp_pk PRIMARY KEY (emp_num),
    CONSTRAINT deptno_positive CHECK (dept_num > 0 AND dept_num < 10),
    CONSTRAINT emp_email_unique UNIQUE (email)
);
```

```
INSERT INTO Employee VALUES (1, 'Faisal', 5, 'faisal@gmail.com');
```

Results:

	emp_num	emp_name	dept_num	email
1	1	Faisal	5	faisal@gmail.com

Lab #03

Introduction to Data Manipulation Language with
Insert ,Update and Delete Command

TASK 1:

- Insert new employee's records.

Code:

```
CREATE TABLE emp(
  empid INT PRIMARY KEY,
  deptno INT FOREIGN KEY (deptno) REFERENCES dept (deptno),
  empname VARCHAR(50),
  empsalary DECIMAL(7,2),
  empprofession VARCHAR(50),
  emphiredate DATE,
);
```

```
INSERT INTO emp VALUES
(1001,50,'Smith',500,'Clerk','2002-04-02'),
(1002,20,'Faisal',5000,'Manager','2001-02-01'),
(1003,50,'Ahmed',900,'Driver','1980-12-21'),
(1004,52,'Farhan',900,'Sales person','1991-10-21'),
(1005,50,'Furqan',2000,'Graphic Designer','1999-11-21')
;
```

Results:

Results		Messages				
	empid	deptno	empname	empsalary	empprofession	emphiredate
1	1001	50	Smith	500.00	Clerk	2002-04-02
2	1002	20	Faisal	5000.00	Manager	2001-02-01
3	1003	50	Ahmed	900.00	Driver	1980-12-21
4	1004	52	Farhan	900.00	Sales person	1991-10-21
5	1005	50	Furqan	2000.00	Graphic Designer	1999-11-21

- Insert new department records deptno=50, dname=ADVERTISING and loc=MIAMI.

Code:

```
CREATE TABLE dept(
  deptno INT PRIMARY KEY,
  deptname VARCHAR(50),
  deptloc VARCHAR(50),
  commission DECIMAL(10, 2)
);
```

```
INSERT INTO dept VALUES
(50,'Advertising','MIAMI',500),
(20,'HR','MIAMI',700),
(52,'Sales','MIAMI',300);
```

Results:

	deptno	deptname	deptloc	commission
1	20	HR	MIAMI	700.00
2	50	Advertising	MIAMI	500.00
3	52	Sales	MIAMI	300.00

- Delete the records of Sales department

Code:

```
DELETE FROM emp WHERE deptno = (SELECT deptno FROM dept WHERE deptname = 'Sales');
DELETE FROM dept WHERE deptname = 'Sales';
```

Results:

	deptno	deptname	deptloc	commission
1	20	HR	MIAMI	700.00
2	50	Advertising	MIAMI	500.00

- Change the employees sal for Smith

Code:

```
UPDATE emp SET empsalary = 600 WHERE empname = 'Smith';
```

Results:

	empid	deptno	empname	empsalary	empprofession	emp hiredate
1	1001	50	Smith	600.00	Clerk	2002-04-02
2	1002	20	Faisal	5000.00	Manager	2001-02-01
3	1003	50	Ahmed	900.00	Driver	1980-12-21
4	1005	50	Furqan	2000.00	Graphic Designer	1999-11-21

- Confirm all these changes by using Select statement

Code:

```
SELECT * FROM emp;
SELECT * FROM dept;
```

Results:

	empid	deptno	empname	empsalary	empprofession	emp hiredate
1	1001	50	Smith	600.00	Clerk	2002-04-02
2	1002	20	Faisal	5000.00	Manager	2001-02-01
3	1003	50	Ahmed	900.00	Driver	1980-12-21
4	1005	50	Furqan	2000.00	Graphic Designer	1999-11-21

	deptno	deptname	deptloc	commission
1	20	HR	MIAMI	700.00
2	50	Advertising	MIAMI	500.00

- Change the salary to \$1,000 for all employees with a salary less than \$900.

Code:

```
update emp set empsalary=1000 where empsalary<900;
```

Results:

	empid	deptno	empname	empsalary	empprofession	emphiredat
1	1001	50	Smith	1000.00	Clerk	2002-04-02
2	1002	20	Faisal	5000.00	Manager	2001-02-01
3	1003	50	Ahmed	900.00	Driver	1980-12-21
4	1005	50	Furqan	2000.00	Graphic Designer	1999-11-21

- Update the salary of each employee to 5000.

Code:

```
update emp set empsalary=5000;
```

Results:

	empid	deptno	empname	empsalary	empprofession	emphiredat
1	1001	50	Smith	5000.00	Clerk	2002-04-02
2	1002	20	Faisal	5000.00	Manager	2001-02-01
3	1003	50	Ahmed	5000.00	Driver	1980-12-21
4	1005	50	Furqan	5000.00	Graphic Designer	1999-11-21

- Change the commission of department 20 to 1000.

Code:

```
UPDATE dept SET commission = 1000 WHERE deptno = 20;
```

Results:

	deptno	deptname	deptloc	commission
1	20	HR	MIAMI	1000.00
2	50	Advertising	MIAMI	500.00

- Change the hire date of all clerks to 02-04-2000

Code:

```
update emp set emphiredat='2000-04-02' where empprofession='Clerk';
```

Results:

	empid	deptno	empname	empsalary	empprofession	emphiredat
1	1001	50	Smith	5000.00	Clerk	2000-04-02
2	1002	20	Faisal	5000.00	Manager	2001-02-01
3	1003	50	Ahmed	5000.00	Driver	1980-12-21
4	1005	50	Furqan	5000.00	Graphic Designer	1999-11-21

- Delete all the records having hire date before 21-dec-81

Code:

```
delete emp where emphiredat<'1981-12-21';
```

Results:

	empid	deptno	empname	empsalary	empprofession	emphiredat
1	1001	50	Smith	5000.00	Clerk	2000-04-02
2	1002	20	Faisal	5000.00	Manager	2001-02-01
3	1005	50	Furqan	5000.00	Graphic Designer	1999-11-21

- Delete all records where salary is greater than 2000.

Code:

```
DELETE FROM emp WHERE empsalary > 2000;
```

Results:

	empid	deptno	empname	empsalary	empprofession	emphiredat
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TASK2:

- Use insert, update & delete commands in 10 different scenarios.

```
CREATE TABLE Student (
  student_id INT PRIMARY KEY,
  student_name VARCHAR(50),
  age INT,
  grade VARCHAR(10),
  address VARCHAR(100));
```

Scenario 1: Inserting a new student record:**Code:**

```
INSERT INTO Student (student_id, student_name, age, grade, address)
VALUES (1, 'Faisal', 20, 'A', 'Burns Road');
```

Results:

Results		Messages			
	student_id	student_name	age	grade	address
1	1	Faisal	20	A	Burns Road

Scenario 2: Updating a student's age:**Code:**

```
UPDATE Student
SET age = 16
WHERE student_id = 1;
```

Results:

Results		Messages			
	student_id	student_name	age	grade	address
1	1	Faisal	16	A	Burns Road

Scenario 3: Deleting a student record:**Code:**

```
DELETE FROM Student
WHERE student_id = 1;
```

Results:

Results		Messages			
	student_id	student_name	age	grade	address

Scenario 4: Inserting multiple student records:**Code:**

```
INSERT INTO Student VALUES
(2, 'Muhammad Faisal', 19, 'A', 'Bahadurabad'),
(3, 'Rehan', 16, 'B', 'Malir Cantt'),
(4, 'Farhan', 12, 'B+', 'Nipa'),
(5, 'Furqan', 19, 'A', 'Gulshan'),
(6, 'Abdul', 16, 'B', 'Bahria Town');
```

Results:

	student_id	student_name	age	grade	address
1	2	Muhammad Faisal	19	A	Bahadurabad
2	3	Rehan	16	B	Malir Cantt
3	4	Farhan	12	B+	Nipa
4	5	Furqan	19	A	Gulshan
5	6	Abdul	16	B	Bahria Town

Scenario 5: Updating a student's grade:**Code:**

```
UPDATE Student
SET grade = 'A+'
WHERE student_name = 'Muhammad Faisal';
```

Results:

	student_id	student_name	age	grade	address
1	2	Muhammad Faisal	19	A+	Bahadurabad
2	3	Rehan	16	B	Malir Cantt
3	4	Farhan	12	B+	Nipa
4	5	Furqan	19	A	Gulshan
5	6	Abdul	16	B	Bahria Town

Scenario 6: Deleting all students with a grade of 'B':**Code:**

```
DELETE FROM Student
WHERE grade = 'B';
```

Results:

	student_id	student_name	age	grade	address
1	2	Muhammad Faisal	19	A+	Bahadurabad
2	4	Farhan	12	B+	Nipa
3	5	Furqan	19	A	Gulshan

Scenario 7: Increase the age of all students by 1 year:**Code:**

```
UPDATE Student
SET age = age + 1;
```

Results:

	student_id	student_name	age	grade	address
1	2	Muhammad Faisal	20	A+	Bahadurabad
2	4	Farhan	13	B+	Nipa
3	5	Furqan	20	A	Gulshan

Scenario 8: Updating a student's address:**Code:**

```
UPDATE Student
SET address = 'Malir'
WHERE student_name = 'Farhan';
```

Results:

Results		Messages			
	student_id	student_name	age	grade	address
1	2	Muhammad Faisal	20	A+	Bahadurabad
2	4	Farhan	13	B+	Malir
3	5	Furqan	20	A	Gulshan

Scenario 9: Deleting all students younger than 15:**Code:**

```
DELETE FROM Student
WHERE age < 15;
```

Results:

Results		Messages			
	student_id	student_name	age	grade	address
1	2	Muhammad Faisal	20	A+	Bahadurabad
2	5	Furqan	20	A	Gulshan

Scenario 10: Update the grade of the student with an ID of 5:**Code:**

```
UPDATE Student
SET grade = 'A'
WHERE student_id = 5;
```

Results:

Results		Messages			
	student_id	student_name	age	grade	address
1	2	Muhammad Faisal	20	A+	Bahadurabad
2	5	Furqan	20	B	Gulshan

Lab #04

Data Retrieval Operations Using Select, Relationship Among Tables Considering Constraints.

TASK:

```
create table dept(
    dept_no int primary key,
    dept_name varchar(50),
    dept_loc varchar(50),
);

create table emp(
    emp_no int primary key,
    dept_no int foreign key(dept_no) references dept (dept_no),
    emp_name varchar(50),
    job varchar(50),
    mgr int,
    emp_salary decimal(7,2),
    emp_comm decimal(7,2),
    emp_phone int
);
```

- List all rows of the table emp.

Code:

```
SELECT * FROM emp;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1001	10	King	President	NULL	5000.00	NULL	1234567890
2	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
3	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
4	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
5	1005	20	Scott Ma...	Analyst	7566	3000.00	NULL	1234567894
6	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895
7	1007	30	Turner	Salesm...	7698	1500.00	0.10	1234567896
8	1008	30	James	Clerk	7698	950.00	NULL	1234567897
9	1009	30	Miller	Salesm...	7698	1250.00	0.20	1234567898
10	1010	30	Allen	Salesm...	7698	1600.00	0.30	1234567899
11	1011	30	Ward	Salesm...	7698	1250.00	0.22	1234567810
12	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- List all rows of the dept.

Code:

```
SELECT * FROM dept;
```

Results:

	dept_no	dept_name	dept_loc
1	10	Sales	BLOCK A
2	20	HR	BLOCK B
3	30	Finance	BLOCK C
4	40	IT	BLOCK D

- List all employees' number from emp.

Code:

```
SELECT emp_no FROM emp;
```

Results:

	emp_no
1	1001
2	1002
3	1003
4	1004
5	1005
6	1006
7	1007
8	1008
9	1009
10	1010
11	1011
12	1012

- List all employees name from emp.

Code:

```
SELECT emp_name FROM emp;
```

	emp_name
1	King
2	Jones
3	Blake
4	Clark
5	Scott Martin
6	Ford
7	Tuner
8	James
9	Miller
10	Allen
11	Ward
12	Smith

- List all departments' number from emp.

Code:

```
SELECT dept_no FROM emp;
```

Results:

	dept_no
1	10
2	20
3	30
4	10
5	20
6	20
7	30
8	30
9	30
10	30
11	30
12	10

- Find all employees whose salaries are between 500 and 1500.

Code:

```
SELECT * FROM emp
WHERE emp_salary BETWEEN 500 AND 1500;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1007	30	Tuner	Salesman	7698	1500.00	0.10	1234567896
2	1008	30	James	Clerk	7698	950.00	NULL	1234567897
3	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
4	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810
5	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- Find all employees whose salaries are between 1500 and 2500.

Code:

```
SELECT * FROM emp
WHERE emp_salary BETWEEN 1500 AND 2500;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
2	1007	30	Tuner	Salesman	7698	1500.00	0.10	1234567896
3	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899

- Find all employees whose salaries are between 2600 and 5000.

Code:

```
SELECT * FROM emp
WHERE emp_salary BETWEEN 2600 AND 5000;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1001	10	King	President	NULL	5000.00	NULL	1234567890
2	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
3	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
4	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
5	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895

- Find all employees whose salaries are less than 2000.

Code:

```
SELECT * FROM emp
WHERE emp_salary < 2000;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1007	30	Turner	Salesman	7698	1500.00	0.10	1234567896
2	1008	30	James	Clerk	7698	950.00	NULL	1234567897
3	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
4	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
5	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810
6	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- Find all employees whose salaries are greater than 2000

Code:

```
SELECT * FROM emp
WHERE emp_salary > 2000;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1001	10	King	President	NULL	5000.00	NULL	1234567890
2	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
3	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
4	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
5	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
6	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895

- Find those employees whose mgr are 7902, 7566, 7788.

Code:

```
SELECT * FROM emp WHERE mgr IN (7902, 7566, 7788);
```


Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
2	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895

- Find those employees whose mgr is between 7788 and 7092.

Code:

```
SELECT * FROM emp WHERE mgr BETWEEN 7092 AND 7788;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
2	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895
3	1007	30	Turner	Salesman	7698	1500.00	0.10	1234567896
4	1008	30	James	Clerk	7698	950.00	NULL	1234567897
5	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
6	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
7	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810
8	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- List all employees whose name starts with 's'.

Code:

```
SELECT * FROM emp WHERE emp_name LIKE 'S%';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
2	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- List all employees whose name start with 'a'.

Code:

```
SELECT * FROM emp WHERE emp_name LIKE 'a%';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899

- List all employees having first name as 'scott'.

Code:

```
SELECT * FROM emp WHERE emp_name LIKE 'Scott%';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894

- List all employees having first name as 'smith' or 'king'.

Code:

```
SELECT * FROM emp WHERE emp_name IN ('smith', 'king');
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1001	10	King	President	NULL	5000.00	NULL	1234567890
2	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- List all employees having first letter in their name 'w' or 'k' and they don't belong to Department number 10,20 and 40.

Code:

```
SELECT * FROM emp WHERE (emp_name LIKE 'w%' OR emp_name LIKE 'k%')
AND dept_no NOT IN (10, 20, 40);
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810

- List all employees whose name begin with 'scott' and end with martin.

Code:

```
SELECT * FROM emp WHERE emp_name LIKE 'scott%martin';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894

- List all employees whose name start with 'm' and belong to department number 30.

Code:

```
SELECT * FROM emp WHERE emp_name LIKE 'm%' AND dept_no = 30;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898

- List all employees whose name started with character 'b' and their jobs are manager.

Code:

```
SELECT * FROM emp WHERE emp_name LIKE 'B%' AND job = 'Manager';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892

- List all employees who do not have job manager..

Code:

```
SELECT * FROM emp WHERE job != 'Manager';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1001	10	King	President	NULL	5000.00	NULL	1234567890
2	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
3	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895
4	1007	30	Turner	Salesman	7698	1500.00	0.10	1234567896
5	1008	30	James	Clerk	7698	950.00	NULL	1234567897
6	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
7	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
8	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810
9	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- Find those employees whose job do not start with 'a'.

Code:

```
SELECT * FROM emp WHERE job not like 'A%';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1001	10	King	President	NULL	5000.00	NULL	1234567890
2	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
3	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
4	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
5	1007	30	Turner	Salesman	7698	1500.00	0.10	1234567896
6	1008	30	James	Clerk	7698	950.00	NULL	1234567897
7	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
8	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
9	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810
10	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- Find those employees whose job does not start with 'c'.

Code:

```
SELECT * FROM emp WHERE job not like 'C%';
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1001	10	King	President	NULL	5000.00	NULL	1234567890
2	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
3	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
4	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
5	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
6	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895
7	1007	30	Turner	Salesman	7698	1500.00	0.10	1234567896
8	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
9	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
10	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810

- List of those employees whose mgr is not null.

Code:

```
SELECT * FROM emp WHERE mgr IS NOT NULL;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
2	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
3	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
4	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
5	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895
6	1007	30	Tuner	Salesman	7698	1500.00	0.10	1234567896
7	1008	30	James	Clerk	7698	950.00	NULL	1234567897
8	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
9	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
10	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810

- List of those employees whose jobs are 'manager or clerk' of department number 10.

Code:

```
SELECT * FROM emp WHERE (job = 'manager' OR job = 'clerk') AND dept_no = 10;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
2	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- List of those employees whose jobs are 'analyst' and 'salesman' of department 30.

Code:

```
SELECT * FROM emp WHERE job IN ('analyst', 'salesman') AND dept_no = 30;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1007	30	Tuner	Salesman	7698	1500.00	0.10	1234567896
2	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
3	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
4	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810

- Find all clerks who earn salaries between 1000 and 2000.

Code:

```
SELECT * FROM emp WHERE job = 'Clerk' AND emp_salary between 1000 and 2000;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- Find all managers who earn salaries between 2500 and 3000.

Code:

```
SELECT * FROM emp WHERE job = 'Manager' AND emp_salary between 2500 and 3000;
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
2	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892

- Find all employees who are either clerk or 'manager' and all employees who earn Salaries in the range of 1000 and 2000.

Code:

```
SELECT * FROM emp WHERE (job = 'Manager' or job = 'Clerk')
OR (emp_salary between 1000 and 2000);
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
2	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
3	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
4	1007	30	Tuner	Salesman	7698	1500.00	0.10	1234567896
5	1008	30	James	Clerk	7698	950.00	NULL	1234567897
6	1009	30	Miller	Salesman	7698	1250.00	0.20	1234567898
7	1010	30	Allen	Salesman	7698	1600.00	0.30	1234567899
8	1011	30	Ward	Salesman	7698	1250.00	0.22	1234567810
9	1012	10	Smith	Clerk	7782	1300.00	NULL	1234567811

- Find all employees who are either manager and/or all employees who earn salaries between 2000 and 3000.

Code:

```
SELECT * FROM emp WHERE job = 'Manager' or (emp_salary between 2000 and 3000);
```

Results:

	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1002	20	Jones	Manager	7839	2975.00	NULL	1234567891
2	1003	30	Blake	Manager	7839	2850.00	NULL	1234567892
3	1004	10	Clark	Manager	7839	2450.00	NULL	1234567893
4	1005	20	Scott Martin	Analyst	7566	3000.00	NULL	1234567894
5	1006	20	Ford	Analyst	7566	3000.00	NULL	1234567895

- Find all employees whose salaries are equal to 1500 and jobs are manager or salesmen.

Code:

```
SELECT * FROM emp WHERE emp_salary = 1500 AND (job = 'manager' OR job = 'salesman');
```

Results:

Results		Messages						
	emp_no	dept_no	emp_name	job	mgr	emp_salary	emp_comm	emp_phone
1	1007	30	Tuner	Salesman	7698	1500.00	0.10	1234567896

Lab #05

Data Retrieval Operations using Join operations

TASK 1:

- Consider the following relation

EMPLOYEE (EMP_ID, EMP_NAME, EMP_ADDRESS, SKILL, PROJ-ID).EQUIPMENT (EQP-ID, EMP_ID, EQP-TYPE, PROJECT).

Code:

```
CREATE TABLE employee(
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  emp_address VARCHAR(50),
  skill VARCHAR(50),
  proj_id INT
);

CREATE TABLE equipment(
  eqp_id INT PRIMARY KEY,
  emp_id INT FOREIGN KEY (emp_id) REFERENCES employee(emp_id),
  eqp_type VARCHAR(50),
  project VARCHAR(50)
);

INSERT INTO employee VALUES
  (1001, 'Faisal', 'Burns Road', 'Developer', 10),
  (1002, 'Ahmed', 'Nipa', 'Engineer', 11),
  (1003, 'Ali', 'Malir', 'Designer', 12)
;

INSERT INTO equipment VALUES
  (110, 1001, 'Laptop', 'Mumbai Project'),
  (111, 1002, 'Welding Machine', 'Renovation Project'),
  (112, 1003, 'Laptop', 'Dubai Project')
;
```

Results:

	emp_id	emp_name	emp_address	skill	proj_id
1	1001	Faisal	Burns Road	Developer	10
2	1002	Ahmed	Nipa	Engineer	11
3	1003	Ali	Malir	Designer	12

	eqp_id	emp_id	eqp_type	project
1	110	1001	Laptop	Mumbai Project
2	111	1002	Welding Machine	Renovation Project
3	112	1003	Laptop	Dubai Project

- Find the join of relations **EMPLOYEE** and **EQUIPMENT**.

Code:

```
SELECT * FROM employee
INNER JOIN equipment ON employee.emp_id=equipment.emp_id;
```

Results:

	emp_id	emp_name	emp_address	skill	proj_id	eqp_id	emp_id	eqp_type	project
1	1001	Faisal	Burns Road	Developer	10	110	1001	Laptop	Mumbai Project
2	1002	Ahmed	Nipa	Engineer	11	111	1002	Welding Machine	Renovation Project
3	1003	Ali	Malir	Designer	12	112	1003	Laptop	Dubai Project

- Get all employees for projects using **EQP-TYPE** as a “Welding machine”.

Code:

```
SELECT employee.emp_id,employee.emp_name,equipment.eqp_id,equipment.eqp_type FROM employee
JOIN equipment ON employee.emp_id = equipment.emp_id
WHERE equipment.eqp_type = 'Welding Machine';
```

Results:

	emp_id	emp_name	eqp_id	eqp_type
1	1002	Ahmed	111	Welding Machine

- Get all machines being used at the **Mumbai Project**?

Code:

```
SELECT employee.emp_id,employee.emp_name,equipment.project FROM equipment
JOIN employee ON employee.emp_id = equipment.emp_id
WHERE equipment.project='Mumbai Project';
```

Results:

	emp_id	emp_name	project
1	1001	Faisal	Mumbai Project

- Find all employees of the project using equipment number **110**.

Code:

```
SELECT employee.emp_id,employee.emp_name,equipment.eqp_id FROM equipment
JOIN employee ON employee.emp_id = equipment.emp_id
WHERE equipment.eqp_id=110;
```

Results:

	emp_id	emp_name	eqp_id
1	1001	Faisal	110

Task 2:

- Consider the following relation

SALESMAN (SALESMAN_ID, NAME, CITY, COMMISSION)

CUSTOMER (CUSTOMER_ID, CUST_NAME, CITY, GRADE, SALESMAN_ID) ORDERS(ORD_NO, PURCH_AMT, ORD_DATE, CUSTOMER_ID, SALESMAN_ID).

Code:

```
CREATE TABLE salesman(
  salesman_id INT PRIMARY KEY,
  name VARCHAR(50),
  city VARCHAR(50),
  commission DECIMAL(7,2),
);
```

```
CREATE TABLE customer(
  customer_id INT PRIMARY KEY,
  cust_name VARCHAR(50),
  city VARCHAR(50),
  grade VARCHAR(50),
  salesman_id INT FOREIGN KEY (salesman_id) REFERENCES salesman(salesman_id)
);
```

```
CREATE TABLE orders(
  ord_no INT PRIMARY KEY,
  purch_amt INT,
  order_date date,
  customer_id INT FOREIGN KEY (customer_id) REFERENCES customer(customer_id),
  salesman_id INT FOREIGN KEY (salesman_id) REFERENCES salesman(salesman_id)
);
```

Results:

Results		Messages			
	salesman_id	name	city	commission	
1	1	Faisal	Karachi	100.00	
2	2	Ahmed	Karachi	200.00	
3	3	Ali	Karachi	150.00	
	customer_id	cust_name	city	grade	salesman_id
1	101	ABC Corporation	New York	1	1
2	102	XYZ Industries	Los Angeles	2	2
3	103	LMN Enterprises	Chicago	1	3
4	104	PQR Ltd.	Houston	3	NULL
	ord_no	purch_amt	order_date	customer_id	salesman_id
1	1001	500	2024-04-01	101	1
2	1002	750	2024-04-05	102	2
3	1003	1000	2024-04-10	101	1
4	1004	1200	2024-04-15	103	3
5	1005	800	2024-04-20	104	NULL

- Write a SQL statement to know which salesman are working for which customer.

Code:

```
SELECT s.NAME AS Salesman_Name, c.CUST_NAME AS Customer_Name
FROM SALESMAN s
INNER JOIN CUSTOMER c ON s.SALESMAN_ID = c.SALESMAN_ID;
```

Results:

	Salesman_Name	Customer_Name
1	Faisal	ABC Corporation
2	Ahmed	XYZ Industries
3	Ali	LMN Enterprises

- Write a SQL statement to make a list in ascending order for the customer who work either through a salesman or by own.

Code:

```
SELECT CUST_NAME, CITY FROM CUSTOMER
ORDER BY SALESMAN_ID ASC, CUST_NAME ASC;
```

Results:

	CUST_NAME	CITY
1	PQR Ltd.	Houston
2	ABC Corporation	New York
3	XYZ Industries	Los Angeles
4	LMN Enterprises	Chicago

- Write a SQL statement to make a list in ascending order for the salesmen who work either for one or more customers or not yet joined under any of the customers.

Code:

```
SELECT s.salesman_id, s.name AS Salesman_Name FROM SALESMAN s
LEFT JOIN CUSTOMER c ON s.SALESMAN_ID = c.SALESMAN_ID
ORDER BY s.SALESMAN_ID ASC;
```

Results:

	salesman_id	Salesman_Name
1	1	Faisal
2	2	Ahmed
3	3	Ali

- **Write a SQL statement to make a report with customer name, city, order no. order date, purchase amount for those customers from the existing list who placed one or more orders or which order(s) have been placed by the customer who are not in the list.**

Code:

```
SELECT c.cust_name, c.city, o.ord_no, o.order_date, o.purch_amt
FROM CUSTOMER c
INNER JOIN ORDERS o ON c.customer_id = o.customer_id
```

UNION ALL

```
SELECT oc.cust_name, oc.city, o.ord_no, o.order_date, o.purch_amt
FROM ORDERS o
LEFT JOIN CUSTOMER oc ON o.customer_id = oc.customer_id
WHERE oc.customer_id IS NULL;
```

Results:

Results		Messages			
	cust_name	city	ord_no	order_date	purch_amt
1	ABC Corporation	New York	1001	2024-04-01	500
2	XYZ Industries	Los Angeles	1002	2024-04-05	750
3	ABC Corporation	New York	1003	2024-04-10	1000
4	LMN Enterprises	Chicago	1004	2024-04-15	1200
5	PQR Ltd.	Houston	1005	2024-04-20	800

Lab #06

Single Row Functions & Aggregating Data Using Group Function**TASK:**

```
CREATE TABLE EMP (
  EMPNO INT PRIMARY KEY,
  ENAME VARCHAR(100),
  JOB VARCHAR(50),
  MGR INT,
  HIREDATE DATE,
  SAL DECIMAL(10, 2),
  COMM DECIMAL(10, 2),
  DEPTNO INT FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO)
);
```

```
CREATE TABLE DEPT (
  DEPTNO INT PRIMARY KEY,
  DNAME VARCHAR(50),
  LOC VARCHAR(50)
);
```

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	1001	Farhan	Clerk	7902	1980-12-17	800.00	NULL	20
2	1002	Furqan	Salesman	7698	1981-02-20	1600.00	300.00	30
3	1003	Wariz	Salesman	7698	1981-02-22	1250.00	500.00	30
4	1004	Faisal	Manager	7839	1981-04-02	2975.00	NULL	20
5	1005	Fahad	Manager	7839	1981-05-01	2850.00	NULL	30
6	1006	Raza	Manager	7839	1981-06-09	2450.00	NULL	10

	DEPTNO	DNAME	LOC
1	10	ACCOUNTING	Newark
2	20	RESEARCH	DALLAS
3	30	SALES	CHICAGO
4	40	OPERATIONS	BOSTON

- Group the employees by their salaries.

Code:

```
SELECT SAL, COUNT(*) AS num_employees FROM EMP
GROUP BY SAL;
```

Results:

	SAL	num_employees
1	800.00	1
2	1250.00	1
3	1600.00	1
4	2450.00	1
5	2850.00	1
6	2975.00	1

- List hiredates in descending order.

Code:

```
SELECT HIREDATE FROM EMP
ORDER BY HIREDATE DESC;
```

Results:

	HIREDATE
1	1981-06-09
2	1981-05-01
3	1981-04-02
4	1981-02-22
5	1981-02-20
6	1980-12-17

- List hiredates in ascending order

Code:

```
SELECT HIREDATE FROM EMP
ORDER BY HIREDATE ASC;
```

Results:

	HIREDATE
1	1980-12-17
2	1981-02-20
3	1981-02-22
4	1981-04-02
5	1981-05-01
6	1981-06-09

- List all employees and add 20 rupees in each salary.

Code:

```
SELECT EMPNO, ENAME, SAL, SAL + 20 AS INCREMENT FROM EMP;
```

Results:

	EMPNO	ENAME	SAL	INCREMENT
1	1001	Farhan	800.00	820.00
2	1002	Furqan	1600.00	1620.00
3	1003	Wariz	1250.00	1270.00
4	1004	Faisal	2975.00	2995.00
5	1005	Fahad	2850.00	2870.00
6	1006	Raza	2450.00	2470.00

- List all 'manager' and add 100 rupees in each salary.

Code:

```
SELECT EMPNO, ENAME, SAL, SAL + 100 AS INCREMENT FROM EMP
WHERE JOB = 'Manager';
```

Results:

	EMPNO	ENAME	SAL	INCREMENT
1	1004	Faisal	2975.00	3075.00
2	1005	Fahad	2850.00	2950.00
3	1006	Raza	2450.00	2550.00

- List all 'salesman and add 500 rupees in each salary.

Code:

```
SELECT EMPNO, ENAME, SAL, SAL + 500 AS INCREMENT FROM EMP
WHERE JOB = 'Salesman';
```

Results:

	EMPNO	ENAME	SAL	INCREMENT
1	1002	Furqan	1600.00	2100.00
2	1003	Wariz	1250.00	1750.00

- List all 'clerk' and add 50 rupees in each salary.

Code:

```
SELECT EMPNO, ENAME, SAL, SAL + 50 AS INCREMENT FROM EMP
WHERE JOB = 'Clerk';
```

Results:

	EMPNO	ENAME	SAL	INCREMENT
1	1001	Farhan	800.00	850.00

- Find eight percent of salesmen salary.

Code:

```
SELECT ENAME, SAL * 0.08 AS eight_percent_salary FROM EMP
WHERE JOB = 'Salesman';
```

Results:

	ENAME	eight_percent_salary
1	Furqan	128.0000
2	Wariz	100.0000

- Find the annual salary of each employee.

Code:

```
SELECT EMPNO, ENAME, SAL * 12 AS ANNUAL_SALARY FROM EMP;
```

Results:

	EMPNO	ENAME	ANNUAL_SALARY
1	1001	Farhan	9600.00
2	1002	Furqan	19200.00
3	1003	Wariz	15000.00
4	1004	Faisal	35700.00
5	1005	Fahad	34200.00
6	1006	Raza	29400.00

- Find the six-month salary of each employee.

Code:

```
SELECT EMPNO, ENAME, SAL * 6 AS SIX_MONTH_SALARY FROM EMP;
```

Results:

	EMPNO	ENAME	SIX_MONTH_SALARY
1	1001	Farhan	4800.00
2	1002	Furqan	9600.00
3	1003	Wariz	7500.00
4	1004	Faisal	17850.00
5	1005	Fahad	17100.00
6	1006	Raza	14700.00

- Find the two-month salary of each manager.

Code:

```
SELECT EMPNO, ENAME, SAL * 2 AS TWO_MONTH_SALARY FROM EMP
WHERE JOB = 'Manager';
```

Results:

	EMPNO	ENAME	TWO_MONTH_SALARY
1	1004	Faisal	5950.00
2	1005	Fahad	5700.00
3	1006	Raza	4900.00

- Make a query in which all the arithmetic expressions will include all the result will be remain same.

Code:

```
SELECT EMPNO, ENAME, SAL * 1 AS SALARY FROM EMP;
```

Results:

	EMPNO	ENAME	SALARY
1	1001	Farhan	800.00
2	1002	Furqan	1600.00
3	1003	Wariz	1250.00
4	1004	Faisal	2975.00
5	1005	Fahad	2850.00
6	1006	Raza	2450.00

- Find the daily wages of each employee.

Code:

```
SELECT EMPNO, ENAME, SAL / 30 AS DAILY_WAGES FROM EMP;
```

Results:

	EMPNO	ENAME	DAILY_WAGES
1	1001	Farhan	26.666666
2	1002	Furqan	53.333333
3	1003	Wariz	41.666666
4	1004	Faisal	99.166666
5	1005	Fahad	95.000000
6	1006	Raza	81.666666

- Find the daily wages of each 'manager'.

Code:

```
SELECT EMPNO, ENAME, SAL / 30 AS DAILY_WAGES FROM EMP
WHERE JOB = 'Manager';
```

Results:

	EMPNO	ENAME	DAILY_WAGES
1	1004	Faisal	99.166666
2	1005	Fahad	95.000000
3	1006	Raza	81.666666

- Display your name in lowercase.

Code:

```
SELECT LOWER('MUHAMMAD FAISAL') AS lower_case_name;
```

Results:

	lower_case_name
1	muhammad faisal

- Find the first and second characters of enames.

Code:

```
SELECT SUBSTRING(ENAME, 1, 2) AS first_two_characters FROM EMP;
```

Results:

	first_two_characters
1	Fa
2	Fu
3	Wa
4	Fa
5	Fa
6	Ra

- Find the minimum salary from the table emp.

Code:

```
SELECT MIN(SAL) AS min_salary FROM EMP;
```

Results:

	min_salary
1	800.00

- Find the maximum salary from the table emp.

Code:

```
SELECT MAX(SAL) AS max_salary FROM EMP;
```

Results:

	max_salary
1	2975.00

- Find the length of all ename.

Code:

```
SELECT ENAME, LEN(ENAME) AS ename_length FROM EMP;
```


Results:

	ENAME	ename_length
1	Farhan	6
2	Furqan	6
3	Wariz	5
4	Faisal	6
5	Fahad	5
6	Raza	4

- Find the length of job.

Code:

```
SELECT JOB, LEN(JOB) AS job_length FROM EMP;
```

Results:

	JOB	job_length
1	Clerk	5
2	Salesman	8
3	Salesman	8
4	Manager	7
5	Manager	7
6	Manager	7

- Find the sum of all salaries.

Code:

```
SELECT SUM(SAL) AS total_salary FROM EMP;
```

Results:

	total_salary
1	11925.00

- Find those employees whose department location is Newark.

Code:

```
SELECT * FROM EMP
WHERE DEPTNO = (
    SELECT DEPTNO FROM DEPT
    WHERE LOC = 'Newark'
);
```

Results:

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	1006	Raza	Manager	7839	1981-06-09	2450.00	NULL	10

- Find those employees who are working in accounting department.

Code:

```
SELECT * FROM EMP
WHERE DEPTNO = (
    SELECT DEPTNO FROM DEPT
    WHERE DNAME = 'ACCOUNTING'
);
```

Results:

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
1	1006	Raza	Manager	7839	1981-06-09	2450.00	NULL	10

- Count all employees.

Code:

```
SELECT COUNT(*) AS num_employees FROM EMP;
```

Results:

	num_employees
1	6

- Display the sum of all employees' salaries.

Code:

```
SELECT SUM(SAL) AS total_salary FROM EMP;
```

Results:

	total_salary
1	11925.00

- How many managers do we have?

Code:

```
SELECT COUNT(*) AS num_managers FROM EMP
WHERE JOB = 'Manager';
```

Results:

	num_managers
1	3

- How many departments do we have?

Code:

```
SELECT COUNT(DISTINCT DEPTNO) AS num_departments FROM EMP;
```

Results:

	num_departments
1	3

- List average salary of each job.

Code:

```
SELECT JOB, AVG(SAL) AS avg_salary FROM EMP
GROUP BY JOB;
```

Results:

	JOB	avg_salary
1	Clerk	800.000000
2	Manager	2758.333333
3	Salesman	1425.000000

- Find the Maximum and Minimum salary of all employees.

Code:

```
SELECT MAX(SAL) AS max_salary, MIN(SAL) AS min_salary FROM EMP;
```

Results:

	max_salary	min_salary
1	2975.00	800.00

- Find the average salaries of those employees who work in dept 10.

Code:

```
SELECT AVG(SAL) AS avg_salary_dept_10 FROM EMP
WHERE DEPTNO = 10;
```

Results:

	avg_salary_dept_10
1	2450.000000

- Find average and sum of all the salaries of each job excluding clerks.

Code:

```
SELECT JOB, AVG(SAL) AS avg_salary, SUM(SAL) AS sum_salary FROM EMP
WHERE JOB != 'Clerk'
GROUP BY JOB;
```

Results:

	JOB	avg_salary	sum_salary
1	Manager	2758.333333	8275.00
2	Salesman	1425.000000	2850.00

- Find the minimum and average salary of each department excluding deptno 10.

Code:

```
SELECT DEPTNO, MIN(SAL) AS min_salary, AVG(SAL) AS avg_salary FROM EMP
WHERE DEPTNO != 10
GROUP BY DEPTNO;
```

Results:

	DEPTNO	min_salary	avg_salary
1	20	800.00	1887.500000
2	30	1250.00	1900.000000

Lab #07**Implementing Cardinalities (minimum, maximum) and Mapping Constraints on Multiple Tables****TASK:**

Daraz is a popular shopping portal and all of us are familiar with its working mechanism . Implement above relations on to following tables of Daraz .

Orders

Product

Seller

Customer

Code:

```
CREATE TABLE Customer (  
    customer_id INT PRIMARY KEY,  
    cust_name VARCHAR(50),  
    cust_address VARCHAR(50),  
    cust_city VARCHAR(50)  
);
```

```
CREATE TABLE Seller (  
    seller_id INT PRIMARY KEY,  
    seller_name VARCHAR(50),  
    seller_address VARCHAR(50),  
    seller_city VARCHAR(50)  
);
```

```
CREATE TABLE Product (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(50),  
    price DECIMAL(10, 2),  
    p_description VARCHAR(50),  
    seller_id INT FOREIGN KEY(seller_id) REFERENCES Seller(seller_id),  
);
```

```
CREATE TABLE Orders (  
    order_id INT PRIMARY KEY,  
    purch_amt DECIMAL(10, 2),  
    order_date DATE,  
    cart_id INT FOREIGN KEY(cart_id) REFERENCES Cart(cart_id),  
);
```

```
CREATE TABLE CART(  
    cart_id INT PRIMARY KEY,  
    product_id INT FOREIGN KEY(product_id) REFERENCES product(product_id),  
    customer_id INT FOREIGN KEY(customer_id) REFERENCES Customer(customer_id)  
);
```

Results:

	customer_id	cust_name	cust_address	cust_city
1	1	Faisal	Burns Road	Karachi
2	2	Farhan	Malir Cantt	Lahore
3	3	Khizer	Defence	Islamabad

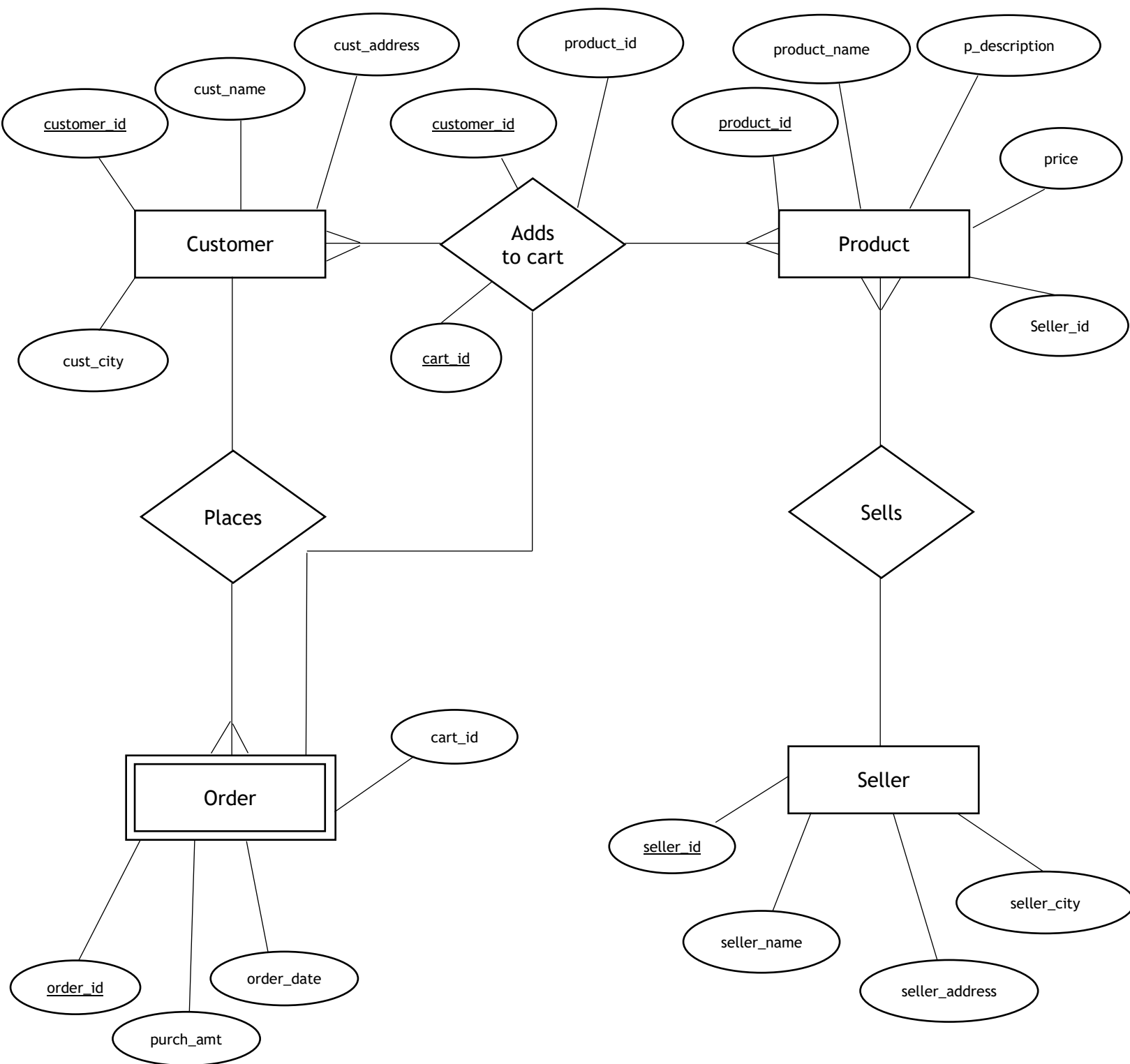
	seller_id	seller_name	seller_address	seller_city
1	101	Raza	Tower	Karachi
2	102	Daniyal	Zainab Market	Karachi
3	103	Anas	Korangi	Karachi

	order_id	purch_amt	order_date	cart_id
1	301	500.00	2024-01-15	501
2	302	21000.00	2024-02-20	502
3	303	20000.00	2024-03-25	503

	product_id	product_name	price	p_description	seller_id
1	201	Smartphone	20000.00	Redmi 9C	101
2	202	T Shirt	500.00	Cotton T-shirt	102
3	203	Smartphone	21000.00	Realme C30	103

	cart_id	product_id	customer_id
1	501	202	1
2	502	203	2
3	503	201	3

ER Diagram



Lab #08

Sub-Queries In Sql**TASK:**

```

CREATE TABLE EMP (
    EMPNO INT PRIMARY KEY,
    ENAME VARCHAR(50),
    JOB VARCHAR(50),
    MGR INT,
    HIREDATE DATE,
    SAL DECIMAL(10, 2),
    DEPTNO INT FOREIGN KEY (DEPTNO) REFERENCES DEPT(DEPTNO)
);
CREATE TABLE DEPT (
    DEPTNO INT PRIMARY KEY,
    DNAME VARCHAR(50),
    LOC VARCHAR(50)
);

```

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	DEPTNO
1	7369	Farhan	Developer	7566	1981-06-09	2450.00	10
2	7499	Talha	Salesman	7698	1981-02-20	1600.00	30
3	7566	Faisal	Manager	7839	1981-04-02	2975.00	20
4	7698	Blake	Manager	7839	1981-05-01	2850.00	30
5	7839	King	President	NULL	1981-11-17	5000.00	10

	DEPTNO	DNAME	LOC
1	10	ACCOUNTING	NEW YORK
2	20	RESEARCH	DALLAS
3	30	SALES	CHICAGO

- Write a query to display the employee name and hiredate for all employees in the same department as Blake. Exclude Blake.

Code:

```

SELECT EMPNO, HIREDATE FROM EMP
  where DEPTNO=(SELECT DEPTNO FROM EMP WHERE ENAME='Blake')
  AND ENAME != 'Blake';

```

Results:

	EMPNO	HIREDATE
1	7499	1981-02-20

- Create a query to display the employee number and name for all employees who earn more than the average salary.

Code:

```

SELECT EMPNO, ENAME, SAL FROM EMP
WHERE SAL > (SELECT AVG(SAL) from EMP)

```

Results:

	EMPNO	ENAME	SAL
1	7839	King	5000.00

- Write a query to display the employee number and name for all employees who work in a department with any employee whose name contains a T.

Code:

```
SELECT EMPNO, ENAME
FROM EMP WHERE DEPTNO IN (
    SELECT DISTINCT DEPTNO FROM EMP
    WHERE ENAME LIKE '%T%'
);
```

Results:

	EMPNO	ENAME
1	7499	Talha
2	7698	Blake

- Display the employee name, department number, and job title for all employees whose department location is Dallas.

Code:

```
SELECT EMPNO, DEPTNO, JOB FROM EMP
WHERE DEPTNO=(SELECT DEPTNO FROM DEPT
    WHERE LOC='DALLAS');
```

Results:

	EMPNO	DEPTNO	JOB
1	7566	20	Manager

- Display the employee name and salary of all employees who report to King.

Code:

```
SELECT ENAME, SAL FROM EMP
WHERE MGR=(SELECT EMPNO FROM EMP
    WHERE ENAME='King');
```

Results:

	ENAME	SAL
1	Faisal	2975.00
2	Blake	2850.00

- Write a query to display the employee name, salary, deptno and job for all employees in the same job as empno 7369.

Code:

```
SELECT ENAME, SAL, DEPTNO, JOB FROM EMP
WHERE JOB=(SELECT JOB FROM EMP
    WHERE EMPNO=7369);
```

Results:

	ENAME	SAL	DEPTNO	JOB
1	Farhan	2450.00	10	Developer

- Display the employee number, name and salary for all employees who earn more than the average salary and who work in department with any employee with a T in their name.

Code:

```
SELECT EMPNO,ENAME, SAL FROM EMP
WHERE SAL>(SELECT AVG(SAL) FROM EMP)
AND DEPTNO IN (
    SELECT DISTINCT DEPTNO
    FROM EMP
    WHERE ENAME LIKE '%T%'
);
```

Results:

Results				Messages			
	EMPNO	ENAME	SAL				

Lab #09

Retrieving Data from Multiple Tables, Mapping of shared data SQL**TASK:**

```

CREATE TABLE departments (
    deptno INT PRIMARY KEY,
    dname VARCHAR(50),
    loc VARCHAR(50)
);
CREATE TABLE employees (
    empno INT PRIMARY KEY,
    ename VARCHAR(50),
    job VARCHAR(50),
    mgr INT,
    hiredate DATE,
    sal DECIMAL(10, 2),
    comm DECIMAL(10, 2),
    deptno INT,
    FOREIGN KEY (deptno) REFERENCES departments(deptno)
);

```

	empno	ename	job	mgr	hiredate	sal	comm	deptno
1	7499	Faisal	Salesman	7566	1981-02-20	1600.00	300.00	30
2	7521	Ahmed	Salesman	7566	1981-02-22	1250.00	500.00	30
3	7566	Farhan	Manager	7839	1981-04-02	2975.00	NULL	20
4	7698	Furqan	Manager	7839	1981-05-01	2850.00	NULL	30
5	7839	King	President	NULL	1981-11-17	5000.00	NULL	10
6	7876	Raza	Analyst	7698	1983-01-12	1100.00	NULL	20

	deptno	dname	loc
1	10	Accounting	Dallas
2	20	Research	New York
3	30	Sales	Chicago

- To display the employee name, department name, and location of all employees who earn a commission.

Code:

```

SELECT E.ename, D.dname, D.loc
FROM employees E, departments D
WHERE E.deptno=D.deptno AND E.comm IS NOT NULL;

```

Results:

	ename	dname	loc
1	Faisal	Sales	Chicago
2	Ahmed	Sales	Chicago

- To display all the employee's name (including KING who has no manager) and their manager name

Code:

```

SELECT emp.ename AS employee_name, mgr.ename AS manager_name
FROM employees emp
LEFT JOIN employees mgr ON emp.mgr = mgr.empno;

```

Results:

	employee_name	manager_name
1	Faisal	Farhan
2	Ahmed	Farhan
3	Farhan	King
4	Furqan	King
5	King	NULL
6	Raza	Furqan

- To display the name of all employees whose manager is *KING*.

Code:

```
SELECT emp.ename AS employee_name, mgr.ename AS manager_name
FROM employees emp
JOIN employees mgr ON emp.mgr = mgr.empno
where mgr.ename= 'King';
```

Results:

	employee_name	manager_name
1	Farhan	King
2	Furqan	King

- Write a query to display the name, job, department number and department name for all employees who work in New York

Code:

```
SELECT E.ename AS employee_name, E.job, E.deptno, D.dname AS department_name
FROM employees E
JOIN departments D ON E.deptno = D.deptno
WHERE D.loc = 'New York';
```

Results:

	employee_name	job	deptno	department_name
1	Farhan	Manager	20	Research
2	Raza	Analyst	20	Research

- Display the employee name and employee number along with their manager's name Manager Number. Label the columns Employee, Emp#, Manager, and Manager#, respectively

Code:

```
SELECT emp.ename AS employee, emp.empno AS emp#, mgr.ename AS manager, mgr.empno AS manager#
FROM employees emp
LEFT JOIN employees mgr ON emp.mgr = mgr.empno;
```

Results:

	employee	emp#	manager	manager#
1	Faisal	7499	Farhan	7566
2	Ahmed	7521	Farhan	7566
3	Farhan	7566	King	7839
4	Furqan	7698	King	7839
5	King	7839	NULL	NULL
6	Raza	7876	Furqan	7698

- **Create a unique listing of all jobs that in department 30. Include the location of department 30 in the Output.**

Code:

```
SELECT DISTINCT E.job, D.loc  
FROM employees E  
JOIN departments D ON E.deptno = D.deptno  
WHERE E.deptno = 30;
```

Results:

	job	loc
1	Manager	Chicago
2	Salesman	Chicago